Haolin Chen

cn.holiechen@gmail.com | LinkedIn | Website | Google Scholar
(530)-220-0439 | Mountain View, CA, 94043

Summary

Research Scientist specializing in **Large Language Model (LLM) training, reinforcement learning, and agentic AI systems**. Experience in leading **end-to-end LLM research**, including pre-training, diffusion-based modeling, and reasoning. Expert in `PyTorch, transformers, TRL, veRL`, TPU/GPU training, and large-scale data synthesis.

Work Experience

Head of Research — Pleasanton, CA
actAVA AI — *March 2026 – Present*

Senior Applied Scientist — Palo Alto, CA
Salesforce AI Research — *August 2025 – February 2026*

Applied Scientist — Palo Alto, CA
Salesforce AI Research — *January 2024 – July 2025*

- Lead LLM reasoning research. Developed **LaTRO**, a reinforcement learning-based algorithm to train LLMs to reason during post-training. The algorithm shows at most ∼ **20% performance gain** on downstream math tasks.
- Lead diffusion language model research, in which LLMs are trained using discrete diffusion. Develop **CoDA-1.7B**, a light-weight, fast diffusion language model for coding that has **performance matching 7B models with 4x efficiency**.
- Lead RL data synthesis research, developed **Webscale-RL**, an open-source data synthesis pipeline that creates high quality RL datasets **at the scale of pre-training data**. Open-sourced a **1 million** dataset for RL.
- Developed the **xLAM** series, a foundational LLM family (**#1 on BFCL leaderboard**) for function calling and agentic environments; showcased demo for travel planning using xLAM at ICLR 2025.
- Prototyping LLM agents for: hierarchical planning, sales pitching, and customer service. Developed **multi-turn agent evaluation** systems with **APIGen-MT**, a multi-turn, agentic data synthesis framework.

Data Scientist — Seattle, WA
Outreach — *August 2022 – September 2023*

- Built PySpark-based predictive tree models for opportunity win rates, improving F1 by **10%** through feature optimization.
- Integrated GPT-3.5 with RAG to summarize deal histories, improving sales insights and automation.

Data Scientist Intern — Seattle, WA
Outreach — *July 2021 – Dec 2021*

- Developed a transformer-based ML system for market research and organization clustering. Outperform the existing rule-based baseline by **7x** in precision, and **3x** in F1 score.

Education

University of California, Davis — Davis, CA
Ph.D. in Applied Mathematics. Advisor: Luis Rademacher. — *Sep. 2017 – July 2022*

Nankai University — Tianjin, China
Bachelor of Science in Physics and Bachelor of Science in Mathematics. — *Sep. 2012 – Jun. 2017*

Selected Publications

Full list: [Google Scholar]

Artificial Intelligence

- **Cura 1T: Specialized Model for Agentic Healthcare**. arXiv preprint arXiv:2607.15314.
  Chen, H., Qi, L., Brown, S., Metelski, D., Xia, T., Lee, J., Wang, Q., Riley, K., Wang, F., & Yao, W.
  Paper
- **χ-Bench: Can AI Agents Automate End-to-End, Long-Horizon, Policy-Rich Healthcare Workflows?** arXiv preprint arXiv:2605.16679.
  Chen, H., Metelski, D., Qi, L., Xia, T., Lee, J., Brown, S., Riley, K., Wang, F., et al.
  Paper
- **CoDA: Coding LM via Diffusion Adaptation**. arXiv preprint arXiv:2510.03270.
  Chen, H., Wang, S., Qin, C., Pang, B., Liu, Z., Qiu, J., Zhang, J., Zhou, Y., Chen, Z., Xu, R., Heinecke, S., Savarese, S., Xiong, C., Wang, H., Yao, W.
  GitHub: 58 ★
- **Webscale-RL: Automated Data Pipeline for Scaling RL Data to Pretraining Levels**. ICLR 2026.
  Cen, Z., *Chen, H.*, Wang, S., Liu, Z., Liu, Z., Zhao, D., ... & Yao, W.
  GitHub: 72 ★
- **Language models are hidden reasoners: Unlocking latent reasoning capabilities via self-rewarding**. arXiv:2411.04282
  Chen, H., Feng, Y., Liu, Z., Yao, W., Prabhakar, A., Heinecke, S., ... & Wang, H.
  GitHub: 123 ★
- **xLAM: A Family of Large Action Models to Empower AI Agent Systems**. 2024 arXiv technical report, arXiv:2409.03215.
  Zhang, J., Lan, T., Zhu, M., Liu, Z., Hoang, T., Kokane, S., Yao, W., Tan, J., Prabhakar, A., *Chen, H.*, Liu, Z., ... & Xiong, C.
  Paper | **GitHub**: 601 ★
- **Apigen-mt: Agentic pipeline for multi-turn data generation via simulated agent-human interplay**. NeurIPS 2025 Datasets & Benchmarks Track, to appear
  Prabhakar, A., Liu, Z., Zhu, M., Zhang, J., Awalgaonkar, T., Wang, S., Liu, Z., *Chen, H* ... & Xiong, C. (2025).

Patents

- **Systems and Methods for Generative Language Model Reasoning Process Optimization**. U.S. Patent Application Publication No. US 2026/0093997 A1; pending U.S. patent application, published April 2, 2026.
  Chen, H., Feng, Y., Prabhakar, A., Liu, Z., Yao, W., Ho, R., Mui, L., Savarese, S., Wang, H., & Xiong, C.
  Patent Application

Machine Learning Theory

- **Overcomplete order-3 tensor decomposition, blind deconvolution and Gaussian mixture models**, SIAM Journal on Mathematics of Data Science 2022 4:1, 336-361.
  Haolin Chen, Luis Rademacher.

Machine Learning in Healthcare

- **The Research Landscape of Multiple Endocrine Neoplasia Type 1 (2000–2021): A Bibliometric Analysis.** Frontiers in Medicine, 1007.
  Feng, C., *Chen, H.*, Huang, L., Feng, Y., & Chang, S. (2022).
- **Trends in worldwide research in inflammatory bowel disease over the period 2012–2021: A bibliometric study.** Frontiers in Medicine, 1389.
  Li, K., Feng, C., *Chen, H.*, Feng, Y., & Li, J. (2022).
- **A Bibliometric Analysis of the Landscape of Parathyroid Carcinoma Research Based on the PubMed (2000–2021).** Frontiers in Oncology, 12, 298.
  Feng, C., Tian, C., Huang, L., *Chen, H.*, Feng, Y., & Chang, S. (2022).
- **Exploring the research landscape of the past, present, and future of thyroid nodules.** Frontiers in Medicine, 9.
  Chen, P., Feng, C., Huang, L., *Chen, H.*, Feng, Y., & Chang, S. (2022).

Other Professional Experience

Reviewer

- NeurIPS 2025
- NeurIPS 2026
- Association for Computational Linguistics
- Mathematical Reviews/MathSciNet
- Journal of Computer Science and Technology